ALEX BAKER

Undergraduate Physics Student · Scientific Machine Learning & Computational Astrophysics

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PROFILE

Undergraduate physics student at Université Laval working on machine-learning and Bayesian-inference methods for scientific datasets, primarily in astrophysics. My work focuses on interpretable, uncertainty-aware pipelines — feature attribution, conformal prediction, posterior diagnostics — applied to stellar spectroscopy, galaxy morphology, and cosmological parameter estimation. Co-author on an arXiv preprint on persistent-memory architectures for large language models. Seeking graduate research at the intersection of machine-learning methodology and physical modelling.

EDUCATION

B.Sc. Physics — Université Laval *Sept. 2023 — expected Winter 2027*

Département de physique, de génie physique et d'optique · Québec, QC, Canada

- Focus on scientific computing, machine learning, and their application to astrophysics.
- Course research project on dimensionality reduction and interpretability for stellar spectral analysis using LAMOST DR5 × Gaia DR3 data; presented at the PHY-3500 departmental symposium, Musée de la civilisation, Winter 2026. Supervised by Prof. Antoine Allard.

PUBLICATIONS & PREPRINTS

Lacasse, S., Hatier, J., & Baker, A. (2026). *MOSS: Modular Orchestrated Storage System — A Persistent Memory Architecture for Large Language Models.* arXiv:2607.04391 (preprint).

- Contributed to the design of the orchestration API and developed Python prototypes for persistent-memory workflows.

RESEARCH PROJECTS

AstroSpectro — Interpretable ML Pipeline for Stellar Spectroscopy *Apr. 2025 — present*

github.com/PhD-Brown/AstroSpectro · phd-brown.github.io/AstroSpectro

- Developed an end-to-end interpretable machine-learning pipeline for stellar spectral analysis using 43,000+ LAMOST DR5 spectra cross-matched with Gaia DR3.
- Engineered 183 physics-based spectroscopic descriptors, including Balmer-line features, metallic-line indicators, molecular bands, continuum indices, and line-profile measurements.
- Trained and evaluated XGBoost-based classifiers; explicitly tested the effect of removing instrumental, positional, and potentially leaky metadata (ROC-AUC 0.964, 87% balanced accuracy).
- Applied PCA, UMAP, t-SNE, autoencoders, clustering, and SHAP interpretability to study the structure of the learned spectral feature space.
- SHAP attribution ranked metallicity-sensitive features (Ca II H&K, Mg b) among the dominant predictors, alongside classical Balmer temperature indicators.
- Manuscript in preparation. 85+ experiment runs tracked with Weights & Biases.

AstroVision — Deep Learning for Galaxy Morphological Classification *Apr. 2025 — present*

github.com/PhD-Brown/AstroVision

- Developed a deep-learning pipeline for galaxy morphological classification on Galaxy10 DECaLS (17,736 images, 10 classes), combining DINOv2 visual features, non-parametric morphometrics (CAS, Gini, M20), and SDSS photometry.
- Benchmarked CNN, EfficientNet-B0, DINOv2 probing, fine-tuning, and late-fusion multimodal strategies; best model achieves 86.3% balanced accuracy.
- Integrated Monte Carlo Dropout uncertainty quantification and conformal prediction to identify ambiguous morphology classifications.
- Prepared an A&A-style manuscript draft and a reproducible codebase.

ξ Dark Energy · BAO MCMC — Bayesian Cosmological Inference *Jan. 2025 — present*
github.com/PhD-Brown/xi-dark-energy-bao-mcmc

- Implemented a reproducible Bayesian MCMC pipeline comparing ΛCDM, CPL, and exploratory oscillating dark-energy parameterisations using Pantheon+SH0ES supernovae and DESI DR2 BAO measurements.
- Built likelihood functions for supernova and BAO distance observables, posterior diagnostics, corner plots, and AIC/BIC model comparison.
- Released as `v0.1.0-exploratory` with explicit documentation of modelling limitations, including fixed sound-horizon assumptions, prior-boundary effects, and self-consistency issues in the oscillating model.

TEACHING EXPERIENCE

Teaching Assistant — Classical Mechanics (PHY-0150) *Winter 2025*
Département de physique, de génie physique et d'optique, Université Laval

- Academic support for undergraduate students in classical mechanics.
- Explanation of complex physical concepts and correction of mathematical problems.

TECHNICAL SKILLS

Scientific Computing	Python · NumPy · SciPy · pandas · Astropy · Matplotlib · Plotly
Machine Learning	scikit-learn · XGBoost · PyTorch · SHAP · UMAP · HDBSCAN · DINOv2 · EfficientNet · MC Dropout · Weights & Biases
Astronomy & Cosmology	FITS processing · LAMOST DR5 · Gaia DR3 · SDSS · Pantheon+SH0ES · DESI BAO · emcee · VizieR · CDS XMatch
Development & Tools	Git · GitHub · GitHub Pages · Jupyter · VS Code · CUDA · joblib · Bash/Linux · LaTeX · Docusaurus

SELECTED RESEARCH OUTPUTS

Manuscript in preparation. Baker, A. *Interpretable Machine Learning for Stellar Spectral Classification with LAMOST DR5 and Gaia DR3* (2026).

Course report & symposium contribution. Baker, A., Jean, J., & Caillat, N. *Réduction de dimension et interprétabilité appliquées aux spectres stellaires LAMOST DR5* (2026). Université Laval; supervised by Prof. Antoine Allard.

Research software. Baker, A. *ξ Dark Energy · BAO MCMC: Exploratory Bayesian Inference for Dark-Energy Models* (2025). Release `v0.1.0-exploratory`. github.com/PhD-Brown/xi-dark-energy-bao-mcmc

LANGUAGES

French — Native **English** — Fluent (scientific writing and communication)